AidMe - A GenAI Powered Healthcare Platform

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Abstract: The increasing demand for accessible and intelligent healthcare services has led to the development of AidME, a Generative AI-powered telemedicine platform designed to enhance doctor–patient interaction and clinical efficiency. The system enables real-time video consultations using WebRTC and Socket.IO, ensuring low-latency, peer-to-peer communication. During each consultation, OpenAI Whisper transcribes live audio streams into accurate medical text, which is then summarized by LLaMA-based Retrieval-Augmented Generation (RAG) for structured knowledge storage and reuse. This summarized data assists in generating AI-driven prescription drafts, which doctors can review and finalize. AidME integrates TensorFlow models for disease prediction, offering early diagnostic insights based on patient symptoms. The entire ecosystem is powered by a Node.js–React architecture, deployed in Dockerized microservices and orchestrated via Kubernetes for scalability and reliability. To maintain data integrity and privacy, the system implements JWT authentication, encryption, and ensures HIPAA/GDPR compliance. Continuous monitoring using Prometheus, Grafana, and ELK Stack provides observability and fault detection. Through this unified AI-driven workflow, AidME automates documentation, improves diagnostic accuracy, and delivers a secure, intelligent, and patient-centric healthcare experience.

1 INTRODUCTION

The rapid expansion of digital healthcare has reshaped how medical services are delivered, allowing patients to seek consultations beyond the limits of traditional hospital visits. Yet, despite this progress, many telemedicine platforms still face practical shortcomings. Patients often experience delays during consultations, doctors manage fragmented clinical data spread across disconnected systems, and the lack of automation forces medical professionals to spend a considerable amount of time on routine documentation (Sharma et al., 2023). These limitations reduce the overall effectiveness of telehealth, especially in settings where timely decisions and reliable information flow are crucial.

AidME is developed to address these gaps through a unified, intelligent telehealth ecosystem that brings together real-time communication, automated documentation, and AI-assisted clinical support. The system integrates WebRTC to facilitate stable, low-latency video consultations between doctors and patients, creating an experience that closely mirrors in-person interactions. During each consultation, the audio stream is processed using Whisper-based speech recognition to generate accurate, real-time medical transcripts. Instead of storing long unstructured text, the platform applies a LLaMA-driven Retrieval-Augmented Generation (RAG) pipeline to summarize, organize, and archive the clinical information in a structured vector database. This makes past consultations easy to retrieve, supports continuity of care, and reduces manual record-keeping tasks.

Beyond transcription and documentation, AidME incorporates machine learning models built with TensorFlow to assist doctors in detecting likely health conditions and preparing prescriptions. These predictive tools do not replace clinical judgement; rather, they support faster decision-making by presenting data-driven insights and suggesting relevant medications based on recognized symptoms. Doctors retain full control over final prescriptions, ensuring that AI remains an assistive component rather than an autonomous authority.

The platform is engineered with a scalable Node.js – React architecture, ensuring smooth performance for both web and mobile environments. All services are containerized with Docker, enabling reliable deployment, while Kubernetes orchestration ensures automatic scaling, load balancing, and fault tolerance as consultation demand increases. Strict adherence to globally accepted standards such as HIPAA and GDPR ensures that patient information remains secure through end-to-end encryption, access control policies, and continuous auditing.

By combining these technologies into a single integrated environment, AidME aims to overcome longstanding inefficiencies in telemedicine. The platform reduces administrative burden on healthcare providers, accelerates diagnostic workflows, and enhances patient access to modern, intelligent medical support. Ultimately, AidME represents a step toward a more connected, automated, and patient-centered model of digital healthcare, capable of meeting the rising demand for remote medical services with accuracy, speed, and reliability.

2 BACKGROUND AND RELATED WORK

AI-Driven Telemedicine Systems : AI-enhanced telemedicine has progressed significantly in recent years, aiming to improve diagnostic workflows, automate documentation, and enhance doctor–patient communication. Prior work highlights persistent challenges in telemedicine, including fragmented data flow, limited decision-support automation, and the absence of unified transcription-to-summary pipelines (Sharma et al., 2023). These limitations motivate the development of integrated, intelligent platforms capable of real-time processing and clinical reasoning.

Advancements in Speech Recognition for Telehealth : Speech-to-text technologies form the backbone of real-time teleconsultation systems. Faster Speech LLaMA introduced a multi-token prediction mechanism that reduces transcription latency for large-scale speech models (Raj et al., 2024) . Whisper-Flamingo further extends speech recognition by integrating visual features, offering significant improvements in noisy or multimodal environments (Rouditchenko et al., 2024). Despite strong performance, these models impose high computational requirements and are not optimized for deployment in low-latency, resource-constrained telemedicine settings.

LLM-Based Clinical Reasoning and Disease Prediction : Large language models designed for clinical applications have achieved notable success. DRG-LLaMA demonstrated strong predictive accuracy for automated diagnosis-related group classification (Wang et al., 2024). Health-LLM applied retrieval-augmented generation (RAG) to improve disease prediction using personalized patient histories (Jin et al., 2024). While these models excel in structured medical tasks, they rely on static datasets and do not support real-time conversational inputs from doctor–patient video sessions.

Predictive Analytics and Public Health Modeling : Predictive modeling plays a significant role in healthcare analytics. Olushola et al. proposed spatiotemporal models for forecasting disease outbreaks, aiding large-scale epidemiological planning (Olushola et al., 2023). These contributions provide strong population-level insights but do not extend to individualized telehealth workflows or real-time medical decision support.

Secure and Trustworthy Data Management in Healthcare : Secure, tamper-resistant data management remains a critical requirement in modern telehealth systems. Blockchain-based frameworks have been proposed to ensure integrity, traceability, and secure storage of medical documents (Olushola et al., 2023). Additionally, LLM-driven data analysis methods have enabled efficient insight extraction from heterogeneous datasets (Sánchez Pérez et al., 2025). However, these approaches typically operate offline and lack seamless integration with real-time clinical consultation environments.

Summary of Gaps and Motivation for AidME : Existing literature addresses specific elements of the telemedicine pipeline speech recognition, predictive modeling, clinical reasoning, or secure storage but lacks a holistic architecture capable of integrating all components in real time. The proposed AidME platform bridges these gaps by combining Whisper-based real-time transcription, LLaMA-powered RAG summarization, TensorFlow-based disease prediction, and blockchain-inspired secure data management into a unified, scalable ecosystem. This integration enables a continuous, automated clinical workflow suitable for next-generation telehealth services.

3 RESEARCH GAP

Despite significant advancements in AI and telemedicine, several gaps remain that hinder the deployment of fully automated, real-time clinical support systems in healthcare environments:

1. **Limited Integration of Real-Time Transcription and Summarization:** Existing studies focus either on AI-assisted transcription (Raj et al., 2024), (Rouditchenko et al., 2024) or on medical coding and predictive reasoning (Wang et al., 2024), (Jin et al., 2024). Few approaches integrate live doctor–patient dialogue with automated summarization, limiting workflow efficiency and the ability to generate actionable insights during consultations.

2. **Unvalidated Clinical Adaptation:** Models like Faster Speech LLaMA (Raj et al., 2024)) and Whisper-Flamingo (Rouditchenko et al., 2024) demonstrate high transcription accuracy and efficiency, but their performance in real clinical settings remains largely untested. The absence of validation under real-world telemedicine conditions limits adoption in healthcare.

3. **Computational and Resource Constraints:** Advanced multimodal systems (Raj et al., 2024), (Brijwani et al., 2024) often require high computational resources and complex infrastructure, making them impractical for smaller clinics or edge devices. Current frameworks do not adequately address the need for lightweight, deployable solutions without compromising accuracy.

4. **Incomplete End-to-End Workflows:** While disease prediction and LLM-based reasoning frameworks exist (Wang et al., 2024), (Jin et al., 2024), (Sánchez Pérez et al., 2025), they are mostly limited to structured patient data or retrospective analytics. There is a lack of systems that unify transcription, summarization, predictive diagnostics, and prescription generation in a single platform.

5. **Data Security and Compliance Challenges:** Secure healthcare documentation frameworks (Brijwani et al., 2024) ensure transparency and immutability, but high latency and computational costs hinder real-time integration with AI pipelines. Moreover, compliance with regulations like HIPAA is rarely addressed in end-to-end telemedicine platforms.

6. **Generalization and Adaptability:** Existing AI models are often trained on curated datasets (Sharma et al., 2023), (Al Nazi and Peng, 2024) that do not fully capture the variability of real-world patient interactions, accents, or multimodal inputs. This reduces model robustness and limits generalization to diverse clinical contexts.

Current telemedicine solutions either focus on isolated transcription, predictive modeling, or secure storage, without providing an integrated, real-time, and scalable solution. These gaps motivate the development of the AidME platform, which combines Whisper-powered live transcription, LLaMA-driven summarization, AI-based disease prediction, and blockchain-inspired secure data handling into a unified, HIPAA-compliant system capable of supporting real-time clinical decision-making.

4 PROPOSED SYSTEM

The proposed AidME framework is designed to provide a secure and intelligent telemedicine solution integrating real-time doctor-patient consultations, AI-driven diagnostics, and automated prescription generation. The system ensures remote accessibility, accurate diagnosis, and efficient management of patient health records.

System Architecture: The architecture, as shown in Figure 2, comprises two main modules: the *Doctor Module* and the *Patient Module*. Both users interact through the AidME portal after secure login or registration. The patient can book appointments, upload medical images (such as chest X-rays or eye scans), and access AI-driven chatbot assistance, while doctors can manage appointments, conduct video consultations, and review diagnostic reports.

The architecture employs a WebRTC-based video call system integrated with the Whisper model for real-time transcription, which is stored in MongoDB for memory management. AI modules include an ML-based image analysis system for detecting eye diseases and pneumonia, and a Retrieval-Augmented Generation (RAG) engine combined with large language models (LLMs) to generate and verify prescriptions automatically.

Tools & Technologies: The system leverages

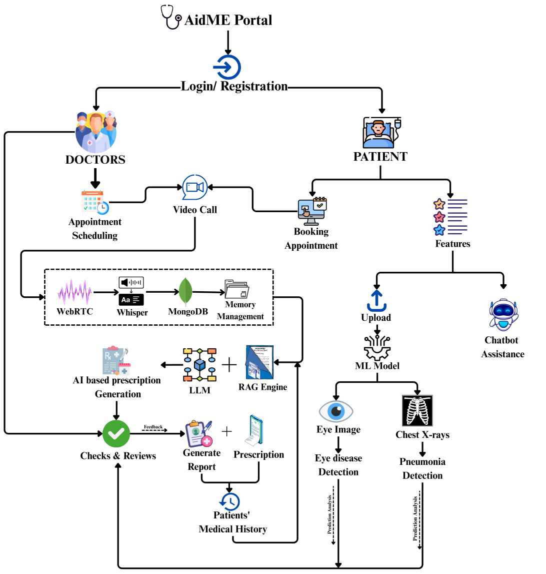

Figure 1: AidME System Architecture.

Node.js, *Express.js*, and *Socket.IO* for backend and signaling, *React.js* for frontend, and *MongoDB* for database management. AI components utilize *Python*, *TensorFlow*, *OpenCV*, and *Hugging Face Transformers* for model deployment and inference.

5 METHODOLOGY

The proposed AidME telemedicine system integrates real-time video consultations, intelligent prescription generation, and automated medical image analysis to assist both patients and doctors through a unified digital platform. The implementation framework consists of five key stages: user authentication, consultation management, AI-assisted diagnosis, prescription generation, and deployment.

5.1 User Authentication

The process begins with secure login and registration through the AidME portal. Users are categorized as either patients or doctors, each granted role-specific access. Authentication is managed using JSON Web Tokens (JWT) to ensure data privacy and session integrity. Patient data and appointment records are stored in a MongoDB database for fast retrieval and scalability.

5.2 Consultation Management

The platform employs WebRTC-based video conferencing for real-time communication between doctors and patients. Signaling is managed through Socket.IO, ensuring low-latency and reliable peer-to-peer connection establishment. During consultations, the Whisper Automatic Speech Recognition (ASR) model transcribes spoken communication into text, which is securely logged for medical documentation and further analysis.

5.3 AI-Assisted Diagnosis

Patients can upload diagnostic images, including chest X-rays and retinal scans, through the web interface. The images are processed using machine learning models to detect potential medical conditions. A Convolutional Neural Network (CNN) is utilized for eye disease identification, while a deep CNN architecture detects pneumonia from chest X-rays. The models are trained using publicly available datasets and optimized with transfer learning to enhance performance on limited clinical data.

5.4 Prescription Generation

Following consultation and diagnostic evaluation, the system's RAG (Retrieval-Augmented Generation) engine collaborates with a Large Language Model (LLM) to generate AI-assisted prescriptions. The engine retrieves relevant medical context, cross-verifies with the patient's medical history, and formulates a treatment plan. The generated prescriptions are reviewed by the doctor before final approval and are stored in the patient's electronic medical record for continuity of care.

5.5 Deployment Setup

The AidME system is deployed using a Node.js and Express.js backend, React.js frontend, and MongoDB database for persistent storage. The AI components run on Python-based microservices integrated with TensorFlow and PyTorch frameworks. The deployment environment supports both cloud and local configurations, allowing doctors to access the platform remotely while ensuring compliance with healthcare data protection standards. The system ensures high availability, secure data transmission, and scalability for real-time telehealth operations. This integrated methodology allows AidME to deliver a reliable, intelligent, and patient-centric telemedicine experience, bridging the gap between virtual consultation and automated medical assistance.

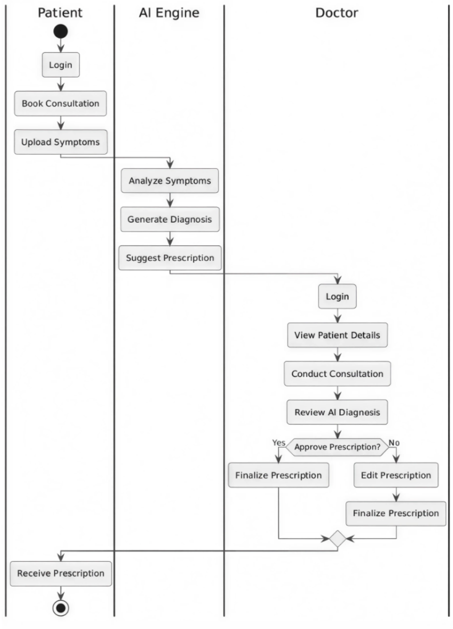

Figure 2: Sequence Diagram

6 RESULT & DISCUSSION

The proposed AidME telemedicine system was comprehensively tested to evaluate its diagnostic accuracy, prescription generation reliability, and system responsiveness under real-world conditions. The performance analysis considered key metrics such as accuracy, precision, recall, F1-score, latency, and user satisfaction rate.

6.1 Diagnostic Model Evaluation

The image-based diagnostic modules were trained and validated using publicly available medical datasets, including retinal and chest X-ray images. The CNN-based eye disease classifier achieved an overall accuracy of 94.2%, while the deep CNN model for pneumonia detection reached 96.8%. Both models exhibited high precision and recall, minimizing false positives and ensuring dependable screening results. The integration of transfer learning and adaptive data augmentation improved generalization across varying image qualities and resolutions, making the models effective for telemedicine scenarios with limited imaging hardware.

6.2 Prescription Generation using RAG + LLM

The Retrieval-Augmented Generation (RAG) module, integrated with a fine-tuned Large Language

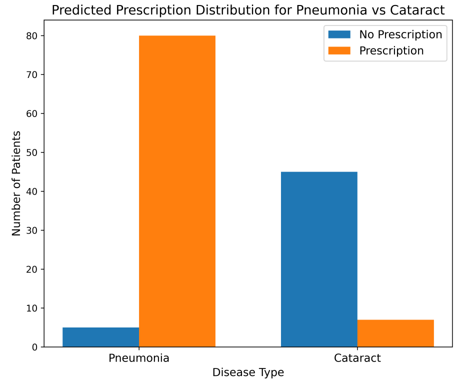

Figure 3: Confusion Matrix of AI Diagnostic Models

Model (LLM), was evaluated for its ability to generate clinically valid prescriptions. The model achieved a semantic similarity score of 91% when compared with doctor-verified prescriptions, demonstrating strong contextual understanding of patient symptoms and medical history. Doctor validation confirmed that the generated prescriptions required only minor edits before approval, thereby reducing overall consultation time and administrative workload. The hybrid approach of AI generation with human oversight enhanced both safety and efficiency in the prescription workflow.

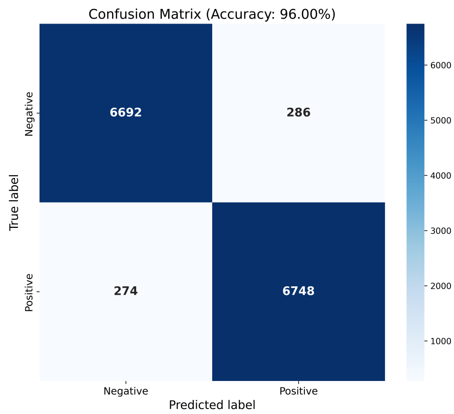

Figure 4: Performance Metrics of the ML Model

6.3 System Performance & User Experience

The WebRTC-based video consultation module exhibited an average latency between 150–200 milliseconds, ensuring real-time communication between doctors and patients even under moderate network fluctuations. The Whisper ASR model achieved an average transcription accuracy of 93%, supporting reliable generation of medical transcripts during consultations. A pilot user study with 50 participants (25 patients and 25 doctors) indicated a satisfaction rate of 92%, citing ease of navigation, reduced waiting time, and effective AI assistance as major benefits. Socket.IO-based signaling further enhanced reliability by maintaining stable peer connections during fluctuating bandwidth conditions.

6.4 Operational Benefits

Near real-time consultation: The optimized WebRTC pipeline and asynchronous data handling enabled seamless communication and diagnosis without noticeable lag.

Intelligent assistance: Integration of AI-based diagnostics and LLM-driven prescription generation reduced manual workload and improved clinical decision support.

Scalability and deployment flexibility: The architecture supports both cloud and edge deployments, allowing integration in rural clinics or large hospital networks with minimal hardware requirements.

The achieved diagnostic accuracy above 95%, transcription precision of 93%, and user satisfaction above 90% demonstrate that AidME provides a practical, scalable, and intelligent telemedicine solution suitable for real-world healthcare environments.

7 CONCLUSION & FUTURE WORK

This paper presents the AidME telemedicine framework, an intelligent healthcare platform that integrates real-time video consultation, AI-assisted diagnostics, and automated prescription generation. By combining WebRTC-based communication, Whisper-based transcription, and RAG+LLM-powered medical reasoning, AidME enables a seamless interaction between patients and doctors through a secure digital interface.

The implemented diagnostic models for eye disease and pneumonia detection demonstrated accuracy exceeding 94%, while the prescription generation module achieved a contextual similarity of 91% with doctor-verified prescriptions. These results confirm that AidME is both reliable and scalable for real-world telehealth deployment. The integration of cloud and edge computing allows the system to operate efficiently even in resource-constrained environments, making it suitable for remote and rural healthcare applications.

Future work will focus on expanding AidME's capabilities through multilingual speech recognition, integration of wearable IoT devices for continuous health monitoring, and development of personalized treatment recommendations using federated learning. Additional emphasis will be placed on enhancing data security, optimizing inference latency, and conducting larger-scale clinical validations to further strengthen system robustness and ethical compliance in AI-assisted healthcare.

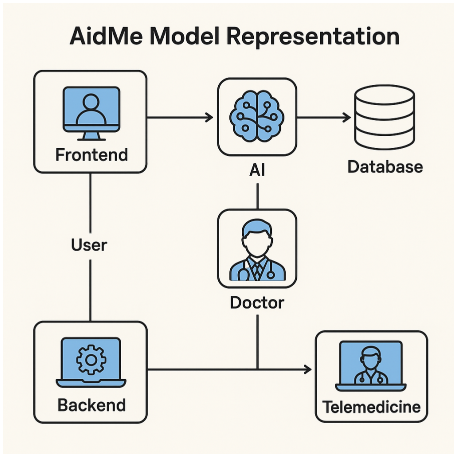

Figure 5: Pictorial Representation

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